

FARSuN Project Overview

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Findability and Accessibility of historical Raw Sunspot Numbers

The International Sunspot Number (ISN) still carries scale jumps such as the 1947 “Waldmeier break”, even after two recalibrations. FARSuN responds by replaying the entire observing chain: **gather sources**, **process data** with Transkribus models and double keying, **validate and standardise** through harmonised observer IDs and calibration curves, and **disseminate** machine-readable products via APIs and VO/EPN-TAP. Legacy corpora like [Wolf’s *Mittheilungen* \(1610–1918\)](#) are now fully integrated with automated QC monitoring NG>NS or metadata gaps, while the complete run of [Zürich observation tables \(1945–1979\)](#) has been digitised with detailed metadata finished for 1945–1950 and observer scaling analyses underway. Dense, spotless-day rich material—[C. H. Adams notebooks \(1819–1823\)](#) and [Gruithuisen manuscripts \(1817–1849\)](#)—is undergoing calibration, orientation, and dual QC before release, and the [Augustin Stark ledgers \(1813–1835\)](#) are being normalised so weather notes align with group counts. Next milestones include expanding uncertainty dashboards, citizen-science transcription workflows, exposing the consolidated database through VO services, and using the harmonised record as the foundation for ISN V3 and long-term solar activity studies.

Data	Description	Time Pe- riod Cov- ered	Importance	Present Status (Early 2025)	Immediate Next Steps
Zürich Tables	Daily counts, Wolf numbers, observing conditions from the Zürich Observatory network.	1945– 1979	–Backbone for late 20th-century sunspot calibration and cross-validation with other observers.	100 % of >2 000 tables digitised; detailed metadata completed for 1945–1950; observer scaling (k-factors) and QC dashboards active.	Finalise metadata for later years, confirm 1980 holdings, publish VO/EPN-TAP access with uncertainty dashboards.
Mittelhelfer Wolf's	compiled sunspot journals extending back to 1610.	1610– 1918	–Provides the longest consolidated series anchoring early historical variability.	Fully integrated in FARSuN schema; automated checks flag NG > NS, duplicates, and missing metadata while harmonisation continues.	Maintain harmonised ob- server/instrument vocabularies and link to new extractions for cross-validation.
C. H. Adams	Full-disk drawings and daily entries highlighting spotless periods.	1819– 1823	–Dense coverage of low-activity years with explicit spotless days for ISN V3 baseline work.	338 drawings (1 056 entries) recovered; direct vs. reflection calibration, orientation, and dewarping underway for positional data.	Complete calibration comparisons with contemporaries, extract coordinates/areas, release with uncertainty notes.
Gruithuisen	Manuscripts in Kurrent script documenting sunspots, weather, and observing context.	1817– 1849	–Bridges early 19th-century observations with detailed metadata valuable for climate links.	Images and text snippets fully collected; two-pass QC (survey + counting) in progress; Transkribus + citizen-science workflows ready.	Finalise daily NG/NS tables, run dual QC, register metadata, and expose FAIR/VO-ready catalogues.

Database	Description	Time Pe- riod Cov- ered	Importance	Present Status (Early 2025)	Immediate Next Steps
Augustin Stark	Added ledgers describing sunspot groups, chains, and contempo- raneous notes.	1813– 1835	–Adds context-rich European observations across Dalton-era variability.	OCR trials successful; layouts normalised and weather annotations linked to counts; integration scripts prepared.	Tighten QC flags, finish linkage to observer registry, and publish summaries aligned with FARSuN datasets.